

Date: 29 May, 2026

To,
Editor

Ref: Paper ID - ICSCSS-685

Dear Editor

We would like to express our gratitude to you and the esteemed reviewers for their thorough and detailed assessment of our manuscript, “**Heterogeneous Multicore SoC Architecture for Real-Time Biomedical Data Analysis**”. Additionally, we would like to express our heartfelt appreciation to your editorial team for providing us with the opportunity to revise the manuscript. We are pleased that the manuscript has been thoroughly reviewed by a panel of distinguished reviewers and that their insightful comments and suggestions have been analyzed.

We made a sincere effort to revise the manuscript in response to the reviewers' concerns. We believe that this revision improved the manuscript's technical merit and readability, for that we are grateful to the reviewers. The detailed point-by-point responses to the reviewers' comments are provided below.

Thanking you,

Yours sincerely,

Reviewer #1

Comment#1: Justify the lack of power consumption analysis for real-time biomedical deployment.

Response: Power Consumption Consideration and Justification

In this paper, no exact physical power consumption data (in milliwatts) are given, because this is a stage where design is at an early implementation phase. The synthesis and functional verification of the system has been carried out on an FPGA platform to show structural feasibility. This initial architectural validation does not include a physical layout tape-out of the complete ASIC with specific technology node libraries to accurately estimate the power consumption of the ASIC. But this low-power operation is inherently guaranteed by a number of important architectural characteristics that are necessary to meet the requirements of portability for biomedical applications: Comparing the number of clock cycles used by offloading the FIR, FFT and CNN operations to dedicated hardware blocks with those used when performing these operations entirely in software on a general processor gives a significant improvement in clock cycles. The fewer active clock cycles translates into lower energy use in the form of dynamics. Single-Cycle Instruction Extensions: The custom VMAC and Bit-Reversal instructions perform complex multi-step mathematical operations in one hardware cycle, rather than consuming valuable power cycles by repeatedly fetching and storing instructions to memory. Minimized Off-Chip Communication: All raw data is continuously streaming, intermediate buffering, and neural network inferences are kept inside the on-chip SRAM, eliminating power-intensive off-chip memory or cloud network transmission, which usually make up a large portion of the energy consumption of wearable medical systems.

Comment#2: Explain how the architecture ensures reliability and fault tolerance in healthcare applications.

Response: These elements result in intermediate FIFO hardware buffering to enhance reliability and fault tolerance for critical healthcare applications, as well as an isolated bus-priority management strategy. The buffering mechanism breaks connection between the various pipeline stages, avoiding any transient delay in processing or computational peaks in

the heavy CNN accelerator from holding up the real-time front- end FIR filter. Moreover, it isolates core control tasks from the accelerators, and under high data loads, the critical hardware components such as the FIR and CNN modules have dedicated memory access, without bus contention or data loss, keeping the integrity of diagnostics unaffected.

Comment#3: It is suggested to address possible bottlenecks caused by shared memory and bus contention.

Response: The architecture features a Time-Division Multiple Access (TDMA) arbitration mechanism on the system bus and localized ping pong buffering in the on-chip SRAM to reduce shared memory bottlenecks and bus contention. The light- weight system bus provides time slots with priorities to the high-throughput accelerators to avoid uncoordinated master requests. At the same time, the dual-banked ping pong buffer design provides simultaneous write access to the FIR filter to one bank of memory while the FFT or CNN is reading an alternate bank of memory. This structural isolation completely removes simultaneous access conflicts to the same memory locations, avoiding bus stalls and maintaining a continuous and real-time data flow.

Comment#4: No research citations like [1], [2], [3] should be given in the conclusion section. Check and remove

Response:

Updated-Conclusion:

In this paper, a heterogeneous RISC-V based System on Chip was presented especially designed for biomedical data analysis in real-time. The architecture consists of a general purpose control core and specialized hardware units for FIR filtering, FFT processing and CNN-based classification, which enables the real time removal of noise and extraction of spectral features on the device, with automated cardiac diagnosis. The system combines use of targeted custom instruction extensions and dedicated hardware acceleration to provide accelerated processing throughput and minimal latency when compared to standard software-only architectures. Besides that, it keeps all calculations within the chip itself, which solidly safeguards patient information privacy by not having to share confidential healthcare data through external networks. Finally, these findings show that an open-source processor architecture, extended with application-specific hardware accelerators, offers an efficient, fast and secure platform for future portable biomedical monitoring and edge healthcare applications.

Comment#5: For the other comments in Section – “Comments 2”:

Proposed heterogeneous SoC functionality and architecture have been evaluated using RTL simulations and FPGA (hardware synthesis). The custom VMAC instruction is used to compute the inner loops of the digital filter in a way that is successful in deterministically calculating the computationally intensive inner loops, as logged in the data stream format (Fig. 4). The proposed hardware-managed FIR architecture is demonstrated to achieve better noise rejection properties and to yield clinically relevant measures of the ECG signal without any phase distortion or latency penalty, and with relatively less sensitivity to high frequency artifacts. The streams are decoupled and the hardware implemented FFT engine uses the custom bit-reversal indexing logic to keep the real-time spectral decomposition. The result shows a high classification accuracy rate of 94.2% for the 8 cardiac classifications, such as complex arrhythmia patterns (PVC and Tachycardia) without transferring any sensitive variables to external networks, proving the ability of the on-chip CNN accelerator to handle the resulting spatial-spectral feature maps correctly. Furthermore, the hardware synthesis metrics (Fig. 6) confirm that the tightly coupled interconnect fabric guarantees no structural hazard, thus verifying that the optimization of the edge-healthcare framework has resulted in a framework with a high performance, high throughput offering custom ISA extensions as well as specialized DSP and neural pipelines.

Metric	Value (%)	Description
Accuracy	94.2	Overall correct classification rate
Precision	93.5	Correctly predicted abnormal cases
Recall	92.8	Ability to detect actual abnormal conditions
F1-Score	93.1	Balance between precision and recall

Processing Stage	Data Type	Data Rate (Approx.)	Bandwidth Requirements
Input Signal	Raw ECG Sample	250-500 samples/sec	Low
FIR Filter	Filtered Signal	Continues Stream	Moderate
FFT Module	Frequency map	Block Based	Moderate
CNN Accelerator	Features Map	Batch Processing	High

Parameter	Cloud Based System	Proposed SoC
Processing Location	Remote Server	On-Chip System
Latency	High	Low and Consistent
Reliability	Network Dependent	Independent of Connectivity